SIMATS ENGINEERING
ASSESSMENT TOOL – 3
Course Name: Digital Image Processing
Course Code: ECA36
COURSE OUTCOME COVERED
CO2: Analyse the various transforms involved in image processing to implement real -time coding with recent software tools. (BL5)
Assessment Tool: Coding Challenge
Prepared by: Dr. K. Thiruppathi

QUESTIONS
Question Bank (10 Questions)
Instructions:
Develop the required program using MATLAB or Python (OpenCV/NumPy).
Display the input and output images with appropriate labels.
Include suitable comments in the program.
Interpret the obtained results wherever applicable.

Q1.
Write a MATLAB/Python program to perform image negative transformation on a grayscale image. Display the original and transformed images and explain the effect of the transformation.

Q2.
Develop a program to perform histogram equalization on a low-contrast image. Plot the histograms before and after enhancement and interpret the improvement in image quality.

Q3.
Write a program to perform contrast stretching using gray-level transformation. Compare the enhanced image with the original image and discuss the changes in contrast.

Q4.
Develop a MATLAB/Python program to implement a 3 × 3 Mean Filter for image smoothing. Compare the filtered image with the original noisy image and comment on the results.

Q5.
Implement a Median Filter to remove salt-and-pepper noise from an image. Compare its performance with the Mean Filter and justify which filter is more suitable.

Q6.
Write a program to implement Laplacian Filtering for image sharpening. Display the sharpened image and explain how edge details are enhanced.

Q7.
Develop a MATLAB/Python program to compute the Two-Dimensional Discrete Fourier Transform (2D-DFT) of an image. Display the magnitude spectrum and explain its significance.

Q8.
Write a program to implement a Low-Pass Filter in the frequency domain. Display the original image, Fourier spectrum, filtered spectrum, and reconstructed image. Interpret the results.

Q9.
Develop a MATLAB/Python program to implement a High-Pass Filter in the

frequency domain for edge enhancement. Compare the output with the original image and discuss the observed changes.

Q10.

A biomedical image contains low contrast and random noise. Develop a MATLAB/Python program to:

Enhance the image using histogram equalization.

Apply an appropriate smoothing filter.

Compare the original and processed images.

Interpret the improvement achieved using suitable image quality observations.

Evaluation Rubric

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

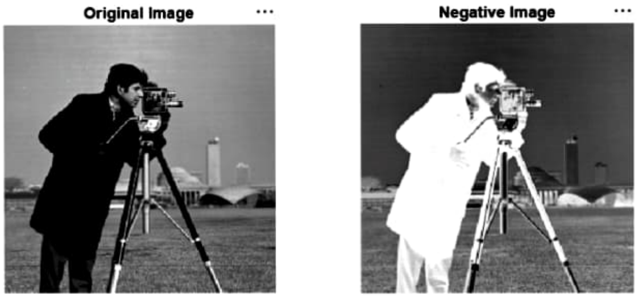
Code of Conduct

I ________________________ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____________________

/MATLAB Drive/histogramequalization.m

```
clc;
clear;
close all;

I = imread('cameraman.tif');

if size(I,3)==3
    I = rgb2gray(I);
end

negative = 255 - I;

figure;
subplot(1,2,1);
imshow(I);
title('Original Image');

subplot(1,2,2);
imshow(negative);
```



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New to MATLAB? See resources for Getting Started.

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LABELS AND ANNOTATIONS

MATLAB Drive

imagenegativetransformation.m | histogramequalization.m | contraststrecting2.m

/MATLAB Drive/contraststrecting2.m

```
clc;
clear;
close all;

I = imread('pout.tif');

if size(I,3)==3
    I = rgb2gray(I);
end

J = imadjust(I,stretchlim(I),[]);

figure;
subplot(1,2,1);
imshow(I);
title('Original');

subplot(1,2,2);
imshow(J);
```

Figure 1

Original

Contrast Stretched

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MATLAB

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MATLAB Drive

imagenegativetransformation.m | histogramequalization.m | contraststrecting2.m | meanfilter2.m

/MATLAB Drive/meanfilter2.m

```
I = imread('cameraman.tif');

noise = imnoise(I,'gaussian',0,0.01);

H = fspecial('average',[3 3]);

filtered = imfilter(noise,H);

figure;
subplot(1,3,1);
imshow(I);
title('Original');

subplot(1,3,2);
imshow(noise);
title('Noisy');

subplot(1,3,3);
```

Figure 1

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LABELS AND ANNOTATIONS

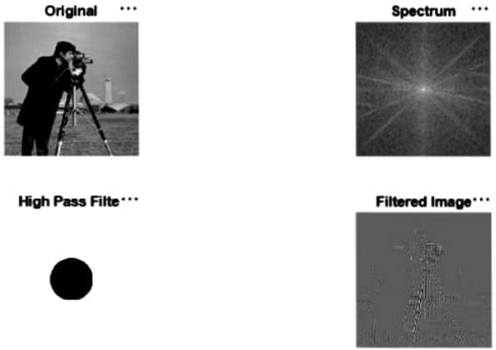
/ > MATLAB Drive

edianfilter.m | laplaciansharpening2.m | Discretetransform2.m | lowpassfilter2.m | highpassfilter2.m

/MATLAB Drive/highpassfilter2.m

```
G = H.*F;

output = real(ifft2(ifftshift(G)));

figure;
subplot(2,2,1);
imshow(I);
title('Original');

subplot(2,2,2);
imshow(log(1+abs(F)),[]);
title('Spectrum');

subplot(2,2,3);
imshow(H,[]);
title('High Pass Filter');

subplot(2,2,4);
```

Figure 1



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LABELS AND ANNOTATIONS

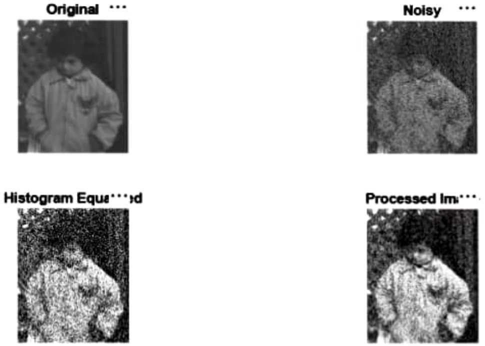
/ > MATLAB Drive

laplaciansharpening2.m × Discretetransform2.m × lowpassfilter2.m × highpassfilter2.m × histogram.m ×

/MATLAB Drive/histogram.m

```
    equalized = histeq(noise);

    smooth = medfilt2(equalized,[3 3]);

    figure;
    subplot(2,2,1);
    imshow(I);
    title('Original');

    subplot(2,2,2);
    imshow(noise);
    title('Noisy');

    subplot(2,2,3);
    imshow(equalized);
    title('Histogram Equalized');

    subplot(2,2,4);
    imshow(smooth);
```

Figure 1 ×



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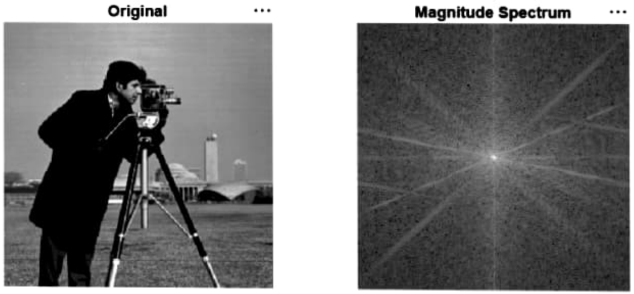
/MATLAB Drive/Discretetransform2.m

```
clc;
clear;
close all;

I = imread('cameraman.tif');

F = fft2(I);

S = fftshift(F);

M = log(1+abs(S));

figure;
subplot(1,2,1);
imshow(I);
title('Original');

subplot(1,2,2);
imshow(M,[]);
```

Figure 1



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LABELS AND ANNOTATIONS

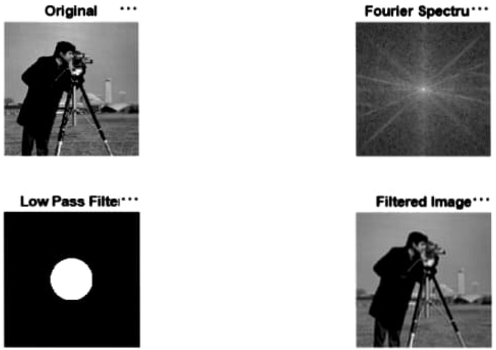
/ > MATLAB Drive

meanfilter2.m | medianfilter.m | laplaciansharpening2.m | Discretetransform2.m | lowpassfilter2.m

/MATLAB Drive/lowpassfilter2.m

```
G = H.*F;

output = real(ifft2(ifftshift(G)));

figure;
subplot(2,2,1);
imshow(I);
title('Original');

subplot(2,2,2);
imshow(log(1+abs(F)),[]);
title('Fourier Spectrum');

subplot(2,2,3);
imshow(H,[]);
title('Low Pass Filter');

subplot(2,2,4);
```

Figure 1



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LABELS AND ANNOTATIONS

/ > MATLAB Drive

gramequalization.m | contraststrecting2.m | meanfilter2.m | medianfilter.m | laplaciansharpening2.m

/MATLAB Drive/laplaciansharpening2.m

```
clc;
clear;
close all;

I = imread('cameraman.tif');

lap = fspecial('laplacian');

filtered = imfilter(double(I),lap);

sharp = uint8(double(I)-filtered);

figure;
subplot(1,2,1);
imshow(I);
title('Original');

subplot(1,2,2);
imshow(sharp);
```

Figure 1

Original

Sharpened

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LABELS AND ANNOTATIONS

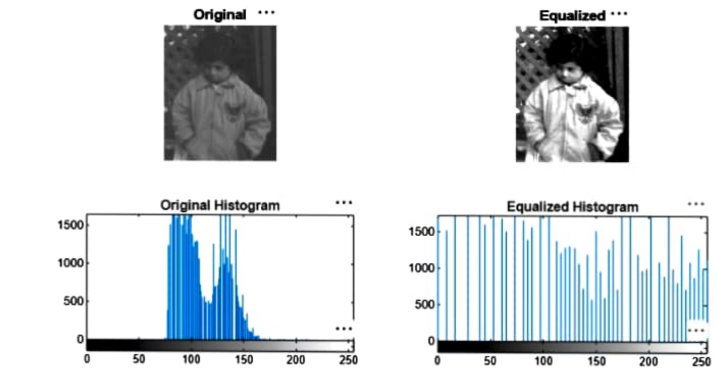
/ > MATLAB Drive

imagenegativetransformation.m

/MATLAB Drive/imagenegativetransformation.m

```
    end

    J = histeq(I);

    figure;
    subplot(2,2,1);
    imshow(I);
    title('Original');

    subplot(2,2,2);
    imshow(J);
    title('Equalized');

    subplot(2,2,3);
    imhist(I);
    title('Original Histogram');

    subplot(2,2,4);
    imhist(J);
```

Figure 1



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LABELS AND ANNOTATIONS

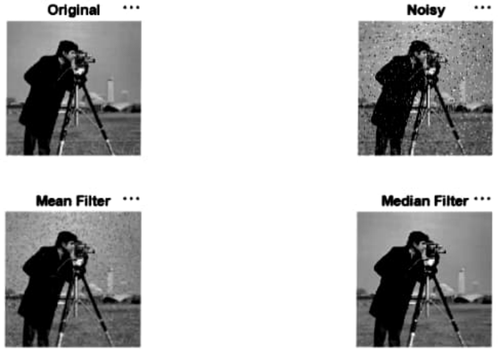
/ › MATLAB Drive

etransformation.m | histogramequalization.m | contraststrecting2.m | meanfilter2.m | medianfilter.m

/MATLAB Drive/medianfilter.m

```
meanFilter = imfilter(noise,fspecial('average',[3 3]));

medianFilter = medfilt2(noise,[3 3]);

figure;
subplot(2,2,1);
imshow(I);
title('Original');

subplot(2,2,2);
imshow(noise);
title('Noisy');

subplot(2,2,3);
imshow(meanFilter);
title('Mean Filter');

subplot(2,2,4);
imshow(medianFilter);
```

Figure 1



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